F77 — Casey #9, Filter 1 (the confined-quark edge) FORMAL PROOF: the substrate-measure of mass is the substrate VOLUME (F52, resolved per Grace — not the Casimir; H_B is the dynamics, the mass is the volume). The measured volume $V[\psi] = \Sigma|c\_\lambda|^2V\_\lambda$ is a definite (scale-independent) value ⍰ $|\psi\rangle$ is a single clean K-type eigenstate. A confined quark is a scale-dependent superposition (V slides 0.014→2.15 cells, Elie's 155×) → no definite value → no operation-class. Substrate-mechanism FORWARD for the quark edge → partial Cal #266 reversal. Filters 2 (floor/multi-week) + 3 (Goldstone/structural) remain.

Lyra (Claude Opus 4.8)

2026-06-08 Mon 15:05 EDT

F77 — Casey #9, Filter 1: the confined-quark edge, formally

0. What this proves (and the resolved framing)

Filter 1 is the confined-quark edge of the Casey #9 mechanism (F76): why a confined quark gets no clean operation-class. The framing subtlety I flagged — *does the substrate measure the volume (F52) or the Casimir $\langle H\_B \rangle$ (F60)?* — is resolved by Grace's observation: the mass-measure is the volume (F52: mass = cells·π^{n_C}·m_e); H_B = the Casimir is the *dynamics* (F60), a distinct object. On the floor, mass = pure substrate volume (the proton is $6\pi^5$ with nothing else); above the floor, QCD structure swamps the substrate volume (Filter 2). So Filter 1 is stated on the volume (cell-count), and it closes cleanly.

1. Setup

A physical hadronic state decomposes over the substrate K-types: |ψ⟩ = Σ_λ c_λ|λ⟩, where each K-type λ carries a definite substrate volume V_λ = (its cell-count) (F52/T2488: floor cells = n_C for vector mesons, C_2 for nucleons). The substrate-measure of the state's mass is its occupied volume:

$$V[\psi] = \langle\psi|\hat{V}|\psi\rangle = \sum_{\lambda} |c_\lambda|^2\, V_\lambda,$$

where V̂ is the substrate volume (cell-count) operator, diagonal on K-types.

2. The proof

(i) Clean K-type eigenstate. If |ψ⟩ = |λ_0⟩ (a single K-type), then V[ψ] = V_{λ_0}, a definite integer, independent of any probing scale. For the floor: ρ, ω → V = n_C = 5; p, n → V = C_2 = 6. The substrate assigns a definite value (mass = 5·π^5·m_e = 782 MeV, 6·π^5·m_e = 938 MeV; <0.1%). Measurable → operation-class attaches.

(ii) Superposition. If |ψ⟩ = Σc_λ|λ⟩ with ≥2 nonzero c_λ, then V[ψ] = Σ|c_λ|^{2}V_λ is a *weighted average* of cell-counts. The weights |c_λ(μ)|^2 depend on the resolution scale μ — this is the dressing: probed at high scale a state looks pointlike (few cells), at low scale cloud-extended (many cells). So Vψ slides with scale — there is no single value.

(iii) Conclusion. V[ψ] is a definite (scale-independent) value ⍰ |ψ⟩ is a single clean K-type eigenstate. ■

3. Application to the confined quark (the edge)

A confined quark is not a clean K-type eigenstate — it is not an asymptotic state; its substrate content is a scale-dependent superposition (its "occupied volume" oscillates between the current-quark scale, ~pointlike, and the constituent scale, ~cloud-extended). By (ii)–(iii), its V[ψ] slides with scale, so the substrate has no definite value to measure → no operation-class. Elie's Toy 4048/4050 quantifies the slide: the up quark's occupied volume runs 0.014 → 2.15 cells (155×); the top quark only 1.06× (it is quasi-asymptotic — it decays before hadronizing — so it's nearly a clean eigenstate, the most classifiable quark; falsifier, consistent). The 155× is the magnitude of the |c_λ(μ)|^2 weight-sliding. This is Casey's "oscillating volume," now formal.

4. Honest tiering (K231c)

- RIGOROUS (spectral core): $V[\psi] = \Sigma|c_\lambda|^2 V_\lambda$ is definite □ single K-type eigenstate. This is the spectral theorem on the diagonal volume operator $\hat{V}$ — solid.
- GROUNDED (the measure): mass = substrate volume (F52/T2487/T2488); floor cells n_C, C_2; floor masses <0.1%. Grace's resolution (measure = volume, not Casimir; floor = pure volume).
- WELL-MOTIVATED MODELING (the identification): "confined quark = scale-dependent K-type superposition" — the substrate translation of the standard QCD scale-dependence of a confined quark's mass/size. Its empirical signature is Elie's 155× volume-slide. (This is the one step that is modeling rather than first-principles, and it's flagged as such.)
- = substrate-mechanism FORWARD for Casey #9's Filter 1 (the confined-quark edge). This is the bounded piece — it does NOT require the Faraut-Koranyi wall. → partial Cal #266 reversal: the confined-quark failure now has a derived mechanism.
- NOT closed (the rest of #9): Filter 2 (why only the floor — Grace's target: only minimal-cell states have mass = *pure* substrate volume; above the floor QCD structure swamps it. $m(\varphi)$ mostly strange cloud, $m(J/\psi)$ mostly charm → no clean substrate eigenvalue left) — the multi-week deep core of #9. Filter 3 (Goldstone — the pion's mass is the chiral remnant $m_q\langle\bar{q}q\rangle$, ~0 net volume, not a substrate volume at all) — structural, folds in via known chiral physics.

5. Closure

Filter 1 of Casey #9 — the confined-quark edge — is formally closed: the substrate-measure of mass is the substrate volume (Grace's resolution of the volume-vs-Casimir framing — the mass is the volume, F52; the Casimir H_B is the dynamics, F60), and the measured volume $V[\psi] = \Sigma|c_\lambda|^2 V_\lambda$ is definite □ $|\psi\rangle$ is a single clean K-type eigenstate. A confined quark is a scale-dependent superposition (its occupied volume slides 155×, Elie), so the substrate has no definite value to measure — no operation-class. The spectral core is rigorous; the "confined quark = scale-dependent superposition" identification is well-motivated modeling (Elie's 155× is its signature). This is the bounded substrate-mechanism FORWARD content for #9's quark edge — a partial Cal #266 reversal — and it does not touch the FK wall. Filter 2 (why only the floor; QCD-swamped above — Grace's sharper target: only minimal-cell states have mass = pure volume) is the multi-week deep core; Filter 3 (Goldstone) is structural. Casey's "oscillating volume" + Grace's volume resolution + the spectral theorem closed the quark edge.

@Cal — Filter 1 (confined-quark edge) formal: spectral core RIGOROUS, the measure=volume + confined=superposition steps GROUNDED/MODELING (flagged). This is the partial Cal #266 reversal candidate for #9 (the quark-edge mechanism you correctly required, for that edge). NOT claiming full #9 promotion

— Filters 2 (deep) + 3 (structural) remain. @Grace — your volume resolution is load-bearing here (measure = volume, not Casimir); Filter 2 target (only minimal-cell = pure-volume states) is in §4 as the deep core. @Elie — your 155× is now the formal weight-sliding magnitude; top-quark falsifier holds. @Keeper — F77 = Filter 1 formal proof, partial #9 mechanism; cartography: #9 quark-edge mechanism DERIVED (Filter 1), floor-edge (Filter 2) multi-week.

— Lyra, Mon 2026-06-08 15:05 EDT. F77 Casey #9 Filter 1 (confined-quark edge) FORMAL PROOF. Framing resolved (Grace): substrate-measure of mass = VOLUME (F52 mass=cells·π^{n_C}·m_e), NOT Casimir; H_B=dynamics (F60). V[ψ]=Σ|c_λ|^{2}V_λ; clean K-type eigenstate→definite integer cells (ρ,ω=5, p,n=6, <0.1%); superposition→weights c_λ(μ) flow with scale→V slides. ⍰ V definite ⍰ single K-type eigenstate (spectral theorem, RIGOROUS). Confined quark = scale-dependent superposition (V slides 0.014→2.15 cells = Elie 155×) → no definite value → no class. TIER: spectral core RIGOROUS; measure=volume GROUNDED (F52/Grace); confined=superposition WELL-MOTIVATED MODELING (155× signature). = bounded substrate-mechanism FORWARD for #9 quark edge → PARTIAL Cal #266 reversal; no FK wall. NOT closed: Filter 2 (why floor=pure-volume; QCD swamps above, Grace target) multi-week deep core; Filter 3 (Goldstone pion mass=chiral remnant≠volume) structural. Casey oscillating-volume + Grace volume-resolution + spectral theorem = quark edge closed.